

Actividad 2

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Actividad 2:
Obtenga los hiperparametros para ajustar la siguiente función lineal $y=ax+by+c$ minimizando la función de costo $J(a,b,c) = \frac{1}{2n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$ utilizando los siguientes datos históricos.

$n=1$
 $n=2$
 $n=3$

	X_1	X_2	y
1	1	1	3
2	2	1	5
3	1	2	4

Solución exacta:
Primera fila
 $a+b+c=3 \Rightarrow c=3-a-b$
Segunda fila
 $2a+b+c=5$
Tercera fila
 $a+2b+c=4$
Resolviendo
 $2a+b+3-a-b=5 \Rightarrow a=2$
 $2+2b+3-2-b=4 \Rightarrow b=1$
 $c=3-2-1=0$
mínimo real: (2, 1, 0)

- Realice dos iteraciones utilizando el Método del gradiente y mida el gap

Sea $\lambda=0.1$ $\hat{y}_i = ax_1 + bx_2 + c$

$$J(a,b,c) = \frac{1}{6} [(3-(a+b+c))^2 + (5-(2a+b+c))^2 + (4-(a+2b+c))^2]$$
$$\frac{\partial J}{\partial a} = \frac{1}{3} [-2(3-(a+b+c)) + 2(5-(2a+b+c))(-2) + 2(4-(a+2b+c))(-1)] = \frac{1}{3} (6a+5b+4c-17)$$
$$\frac{\partial J}{\partial b} = \frac{1}{3} [-2(3-a-b-c) + (-2)(5-2a-b-c) + 2(4-a-2b-c)(2)] = \frac{1}{3} (5a+6b+4c-16)$$
$$\frac{\partial J}{\partial c} = \frac{1}{3} [(-2)(3-a-b-c) + (-2)(5-2a-b-c) + (-1)(4-a-2b-c)] = \frac{1}{3} (4a+4b+3c-12)$$

Primera iteración con $(a_0, b_0, c_0) = (0, 0, 0)$

$$\frac{\partial J}{\partial a} = \frac{1}{3} (6(0) + 5(0) + 4(0) - 17) = -\frac{17}{3}$$
$$\frac{\partial J}{\partial b} = \frac{1}{3} (5(0) + 6(0) + 4(0) - 16) = -\frac{16}{3}$$
$$\frac{\partial J}{\partial c} = \frac{1}{3} (4(0) + 4(0) + 3(0) - 12) = -4$$
$$\nabla J(a_0, b_0, c_0) = \left(-\frac{17}{3}, -\frac{16}{3}, -4 \right)$$
$$a_1 = a_0 - \lambda \frac{\partial J}{\partial a} = 0 - 0.1 \left(-\frac{17}{3} \right) = 0.5667$$
$$b_1 = b_0 - \lambda \frac{\partial J}{\partial b} = 0 - 0.1 \left(-\frac{16}{3} \right) = 0.5333$$
$$c_1 = c_0 - \lambda \frac{\partial J}{\partial c} = 0 - 0.1 (-4) = 0.4$$

Segunda iteración

$$\hat{y}_1 = a+b+c = 0.5667 + 0.5333 + 0.4 = 1.5$$
$$\hat{y}_2 = 2a+b+c = 2(0.5667) + 0.5333 + 0.4 = 2.0067$$
$$\hat{y}_3 = a+2b+c = 0.5667 + 2(0.5333) + 0.4 = 2.0333$$
$$\frac{\partial J}{\partial a} = \frac{1}{3} (6(1.5) + 5(2.0067) + 4(2.0333) - 17) = -3.111$$
$$\frac{\partial J}{\partial b} = \frac{1}{3} (5(1.5) + 6(2.0067) + 4(2.0333) + 16) = -2.7889$$
$$\frac{\partial J}{\partial c} = \frac{1}{3} (4(1.5) + 4(2.0067) + 3(2.0333) - 12) = -2.133$$

$$a_2 = 0.5667 - 0.1(3.111) = 0.8778$$
$$b_2 = 0.5333 - 0.1(-2.7889) = 0.8122$$
$$c_2 = 0.4 - 0.1(-2.133) = 0.6133$$

- Realice dos épocas utilizando SGD y mida el GAP

Época 1: $(a,b,c) = (0,0,0)$ $\nabla J_i = (\hat{y}_i - y_i) \begin{bmatrix} x_1 \\ x_2 \\ 1 \end{bmatrix}$

usando orden de los datos: (1,1,3)

$$\hat{y}_1 = 0 \Rightarrow \hat{y}_1 - y_1 = 0 - 3 = -3$$
$$\nabla J_1 = (-3) \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -3 \\ -3 \\ -3 \end{bmatrix}$$
$$(a,b,c) = (0,0,0) - 0.1(-3,-3,-3)$$
$$(a,b,c) = (0.3, 0.3, 0.3)$$

Ahora (2,1,6)

$$\hat{y}_1 = 2(0.3) + 0.3 + 0.3 = 1.2$$
$$\hat{y}_1 - y_1 = 1.2 - 5 = -3.8$$
$$\nabla J_2 = (-3.8) \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -7.6 \\ -3.8 \\ -3.8 \end{bmatrix}$$
$$(a,b,c) = (0.3, 0.3, 0.3) - 0.1(-7.6, -3.8, -3.8)$$
$$(a,b,c) = (1.06, 0.63, 0.63)$$

Para (1,2,4)

$$\hat{y}_1 = 1.06 + 2(0.63) + 0.63 = 3.10$$
$$\hat{y}_1 - y_1 = 3.1 - 4 = -0.9$$
$$\nabla J_3 = (-0.9) \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -0.9 \\ -1.8 \\ -0.9 \end{bmatrix}$$
$$(a,b,c) = (1.06, 0.63, 0.63) - 0.1(-0.9, -1.8, -0.9)$$
$$(a,b,c) = (1.15, 0.86, 0.77)$$
$$J = 0.2205$$

Época 2

con (1,1,3)

$$\hat{y}_1 = 1.15 + 0.86 + 0.77 = 2.78$$
$$\hat{y}_1 - y_1 = 2.78 - 3 = -0.22$$
$$\nabla J_1 = (-0.22) \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -0.22 \\ -0.22 \\ -0.22 \end{bmatrix}$$
$$(a,b,c) = (1.15, 0.86, 0.77) - 0.1 \nabla J_1$$
$$(a,b,c) = (1.172, 0.882, 0.792)$$

con (2,1,5)

$$\hat{y}_1 = 2(1.172) + 0.882 + 0.792 = 4.018$$
$$\hat{y}_1 - y_1 = 4.018 - 5 = -0.982$$
$$\nabla J_2 = -0.982 \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1.964 \\ -0.982 \\ -0.982 \end{bmatrix}$$
$$(a,b,c) = (1.172, 0.882, 0.792) - 0.1 \nabla J_2$$
$$(a,b,c) = (1.3684, 0.9802, 0.8902)$$

Para (1,2,4)

$$\hat{y}_1 = 1.3684 + 2(0.9802) + 0.8902 = 4.219$$
$$\hat{y}_1 - y_1 = 4.219 - 4 = 0.219$$
$$\nabla J_3 = (0.219) \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.219 \\ 0.438 \\ 0.219 \end{bmatrix}$$
$$(a,b,c) = (1.37, 0.98, 0.89) + 0.1 \nabla J_3$$
$$(a,b,c) = (1.365, 0.9364, 0.8683)$$
$$J(a,b,c) = 0.0471$$

- ¿Qué algoritmo converge más rápido?

Teniendo en cuenta que la solución real es $(a^*, b^*, c^*) = (2, 1, 0)$

Calculamos el gap:

Gradiente: $(a,b,c) = (0.8773, 0.8122, 0.6133)$

$$\hat{y}_1 = 2.3033, \quad \hat{y}_2 = 3.1811, \quad \hat{y}_3 = 3.155$$
$$J = \frac{1}{6} [(3-2.3033)^2 + (5-3.1811)^2 + (4-3.155)^2]$$
$$J = 0.7627$$

SGD: $(a,b,c) = (1.3465, 0.9364, 0.8683)$

$$\hat{y}_1 = 3.1512, \quad \hat{y}_2 = 4.4977, \quad \hat{y}_3 = 4.0876$$
$$J = \frac{1}{6} [(3-3.1512)^2 + (5-4.4977)^2 + (4-4.0876)^2]$$
$$J = 0.0471$$

Como $0.0471 < 0.7627$, SGD está más cerca del mínimo $J^* = 0$, entonces converge más rápido.